

WIRELESS NETWORKING

Day – 4 Hands on subnetting practical.

Task –1 Dividing the IP address in 4 parts.

- To divide the 192.168.10.0/24 network into 4 equal subnets, you need to borrow 2 bits from the host portion .
- Original CIDR: /24
- New Subnet CIDR: /26
- New Subnet Mask: 255.255.255.192
- Block Size (Increment): $256 - 192 = 64$
- Usable Hosts per Subnet: $2^{\{(32-26)\}} - 2 = 62$

Subnet	Subnet Mask	Network Address	Usable Host Range	Broadcast Address
Subnet 1	255.255.255.192 (/26)	192.168.10.0	192.168.10.1 – 192.168.10.62	192.168.10.63
Subnet 2	255.255.255.192 (/26)	192.168.10.64	192.168.10.65 – 192.168.10.126	192.168.10.127
Subnet 3	255.255.255.192 (/26)	192.168.10.128	192.168.10.129 – 192.168.10.190	192.168.10.191
Subnet 4	255.255.255.192 (/26)	192.168.10.192	192.168.10.193 – 192.168.10.254	192.168.10.255

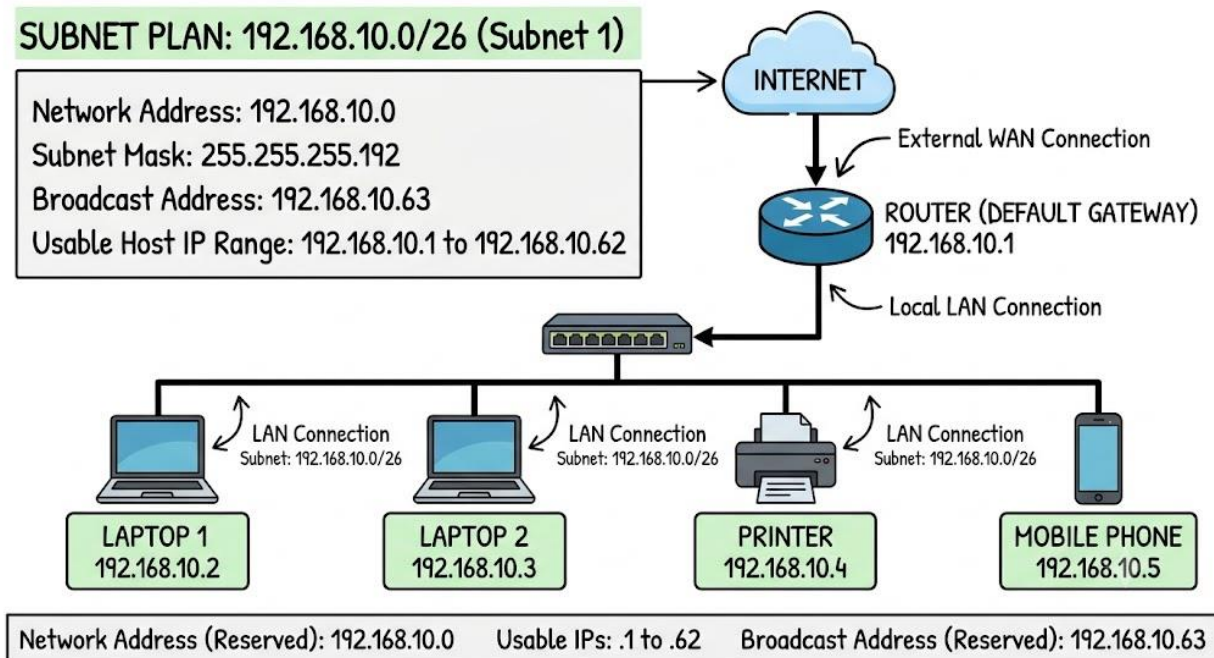
Task –2 Assign the IP address to the devices.

- Subnet 1 Network Address: 192.168.10.0

- Subnet Mask: 255.255.255.192 (/26)
- Broadcast Address: 192.168.10.63
- Usable Host IP Range: 192.168.10.1 to 192.168.10.62
- Device IP Assignments :-
- Router (Default Gateway): 192.168.10.1
- Laptop 1: 192.168.10.2
- Laptop 2: 192.168.10.3
- Printer: 192.168.10.4
- Mobile Phone: 192.168.10.5

Device	Assigned IP Address	Subnet Mask	Default Gateway
Router	192.168.10.1	255.255.255.192	192.168.10.1 (Self)
Laptop 1	192.168.10.2	255.255.255.192	192.168.10.1
Laptop 2	192.168.10.3	255.255.255.192	192.168.10.1
Printer	192.168.10.4	255.255.255.192	192.168.10.1

Task – 3 Labelling the subnet and range.



Task – 4 Calculating the subnets on IP address.

- Step 1: Understand Your Starting Block
- Given Network: 192.168.10.0/24
- The /24 means the first 24 bits out of 32 total bits are locked for the network path.
- That leaves 8 bits ($32 - 24 = 8$) for device (host) IP addresses.
- An 8-bit space gives $2^8 = 256$ total IP addresses (.0 through .255).
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- Step 2: Borrow Bits to Create 4 Subnets
- To split one big network into 4 smaller, equal subnets, we need to calculate how many binary bits to "borrow" from the host portion:
- Formula: $2^{\text{borrowed bits}} = \text{Number of Subnets}$
- To get 4 subnets: $2^2 = 4$, so we borrow 2 bits.
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- Step 3: Calculate the New Subnet Mask
- Add the 2 borrowed bits to the original prefix: $24 + 2 = \text{bf}/26$.
- In binary, a /26 prefix looks like:
- 11111111 . 11111111 . 11111111 . 11000000
- Convert that last binary byte (11000000) into decimal ($128 + 64 = 192$).
- New Subnet Mask: 255.255.255.192
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- Step 4: Find the Block Size (Magic Number)
- The block size tells you where each subnet starts and ends:
- Subtract the last octet of the subnet mask from 256:
- $256 - 192 = \text{bf}64$
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- Your subnets will increment by 64:
- Subnet 1: 192.168.10.0
- Subnet 2: 192.168.10.64
- Subnet 3: 192.168.10.128
- Subnet 4: 192.168.10.192
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- Step 5: Calculate Usable Hosts Per Subnet
- Remaining host bits: $32 - 26 = \text{bf}6\text{ bits}$.
- Total IPs per subnet: $2^6 = 64$.
- Rule: You must subtract 2 addresses from every subnet (1 for the Network Address that names the group, and 1 for the Broadcast Address that sends data to all devices)

- Usable Hosts: $64 - 2 = \mathbf{62}$ devices per subnet\$.
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- Step 6: Assign IPs to Devices (Using Subnet 1)
- For Subnet 1 (192.168.10.0/26):
- Network Address (Off-limits): 192.168.10.0
- Broadcast Address (Off-limits): 192.168.10.63 (the number right before Subnet 2 starts)
- Valid Host Range: 192.168.10.1 to 192.168.10.62
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- Assigning to devices sequentially:
- Router Gateway: 192.168.10.1
- Laptop 1: 192.168.10.2
- Laptop 2: 192.168.10.3
- Printer: 192.168.10.4
- Mobile: 192.168.10.5
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